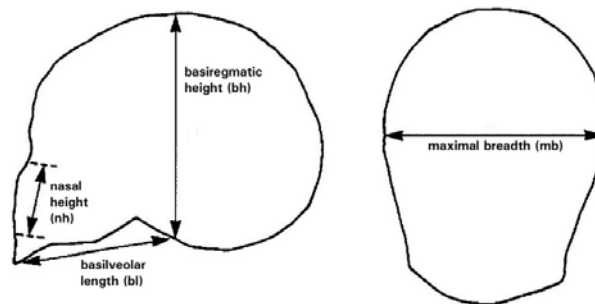


Problem 5.1 (MATLAB) **MANOVA 1**

The file '**dataset_problem_5_1.mat**' contains three 30x4 data matrices containing size measurements for Egyptian skull samples. Each of the three samples is from a specific time period in history (4000 B.C., 3300 B.C. and 1850 B.C., respectively) and contains 30 skull observations for that period.

For each of the skulls are observed four variables:



- X_1 = maximum breadth
- X_2 = basibregmatic height
- X_3 = basalveolar length
- X_4 = nasal height,

with all measurements in unit [mm].

Researchers have suggested that a change in skull size over time is evidence of the interbreeding of a resident population with immigrant population. Therefore it is desired to test for any significant change of skull size between the three time periods.

Preliminary examination of samples, part 1: Bartlett test

First it shall be established whether the sample covariance matrices for the three samples can be pooled, that is to test

$$H_{0,\Sigma}: \Sigma_{\text{time1}} = \Sigma_{\text{time2}} = \Sigma_{\text{time3}}$$

Assuming therefore that

$$X_{\text{time1},i} \sim N_4(\mu_{\text{time1}}, \Sigma_{\text{time1}}), \quad i = 1, 2, \dots, 30$$

$$X_{\text{time2},i} \sim N_4(\mu_{\text{time2}}, \Sigma_{\text{time2}}), \quad i = 1, 2, \dots, 30$$

$$X_{\text{time3},i} \sim N_4(\mu_{\text{time3}}, \Sigma_{\text{time3}}), \quad i = 1, 2, \dots, 30$$

with all parameters unknown

- Load the data matrices
- Calculate the descriptive statistics
- Perform a Bartlett test of $H_{0,\Sigma}$. Use level of significance $\alpha = 0.05$ but calculate also the p-value for the test
- Conclude

Preliminary examination of samples, part 2: check of multivariate normal model

Perform a model check for multivariate normality for each of the three samples:

- Make plots of the multivariate scatter matrices plus marginal histograms
- Calculate the squared sample Mahalanobis distances, $d_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$, and make QQ-plots versus the relevant χ^2 distribution for check of approximate linearity
- Conclude

Test for equality of skull size mean vectors from the three time periods (MANOVA1)

Thus having (hopefully) reasonably well established that

$$X_{\text{time1},i} \sim N_4(\mu_{\text{time1}}, \Sigma), \quad i = 1, 2, \dots, 30$$

$$X_{\text{time2},i} \sim N_4(\mu_{\text{time2}}, \Sigma), \quad i = 1, 2, \dots, 30$$

$$X_{\text{time3},i} \sim N_4(\mu_{\text{time3}}, \Sigma), \quad i = 1, 2, \dots, 30$$

the main purpose of this investigation can be addressed, namely to test for equality of skull size mean vectors from the three time periods, that is to test

$$H_{0,\mu}: \mu_{\text{time1}} = \mu_{\text{time2}} = \mu_{\text{time3}}$$

Perform in three different ways a MANOVA1 for the data from the three time periods

- 1. Exact test:
- Calculate the Sum of Squares decomposition and their degrees of freedom
- Calculate Wilks Lambda and the corresponding test-statistic for the case at hand
- Perform an exact test of the hypothesis $H_{0,\mu}$, using the appropriate F-distribution for calculation of critical value. Use level of significance $\alpha = 0.05$ but calculate also the p-value for the test
- 2. Approximate test:
- Perform a large-sample approximate test of the same hypothesis using the appropriate Chi-squared distribution for calculation of critical value. Use again $\alpha = 0.05$ but calculate also the p-value
- 3. Approximate test using built-command "**manova1**":
- Compare the three test-results
- Conclude the test by estimating the remaining model parameters, which is either (μ, Σ) or $(\mu_{\text{time1}}, \mu_{\text{time2}}, \mu_{\text{time3}}, \Sigma)$ depending on whether you accept or reject $H_{0,\mu}$

Confidence intervals for difference in mean vectors

Turning to confidence intervals, still using $\alpha = 0.05$,

- Calculate pair-wise, simultaneous Bonferroni confidence intervals for difference in mean vector variables between the time periods (3 time period pairs and 4 variables giving a total of 12 pair-wise, simultaneous Bonferroni confidence intervals)
- Inspect the various confidence intervals and conclude
- Compare with the hypothesis test results and explain

Problem 5.2 (MATLAB)

MANOVA 2

The file '**dataset_problem_5_2.mat**' contains measurements on three growth-related variables for peanut plants. A two-factor study is being performed with the intention to improve the yield of peanut farms in USA. Two different locations of farms were used (factor 1), at each location using three different varieties of peanut plants (factor 2). For each of the six combinations of factors 1 and 2, two observations were made on the three variables:

- X_1 = yield [plot weight]
- X_2 = weight of sound mature kernels [grams]
- X_3 = seed size [grams].

The dataset file is of size 12x5, where each row corresponds to an observation and the columns have the following formatting:

- Col 1 = factor 1 (location), value 1 or 2
- Col 2 = factor 2 (variety), value 1, 2 or 3
- Col 3-5 = X_1, X_2, X_3

It is desired to perform a two-way MANOVA for test for any significant location effect, variety effect and location-variety interaction effect.

MANOVA2 model

- $X_{lkr} \sim N_3(\mu_{lk}, \Sigma_{lk}), l = 1, \dots, g = 2, k = 1, \dots, b = 3, r = 1, \dots, n = 2$
- $\forall \Sigma_{lk} = \Sigma$
- $\mu_{lk} = \mu + \tau_l + \beta_k + \gamma_{lk}$

Sums of squares decomposition

- Calculate first the sample means $\hat{\mu}_{lk}, \hat{\mu}_{l\cdot}, \hat{\mu}_{\cdot k}, \hat{\mu}$
- Calculate the SS decomposition $SS_T = SS_{B1} + SS_{B2} + SS_{int} + SS_W$
- Indicate the degrees of freedom for each of the SS's

Test for systematic interaction between location and variety

- Perform a test of the hypothesis $H_{0,\gamma}: \forall \gamma_{lk} = 0$ (no systematic interaction).
Use level of significance $\alpha = 0.01$ but calculate also the p-value for the test

Test for systematic factor 1 effect (location)

- Perform a test of the hypothesis $H_{0,\tau}: \forall \tau_l = 0$ (no location effect).
Use level of significance $\alpha = 0.01$ but calculate also the p-value for the test

Test for systematic factor 2 effect (variety)

- Perform a test of the hypothesis $H_{0,\beta}: \forall \beta_k = 0$ (no variety effect).
Use level of significance $\alpha = 0.01$ but calculate also the p-value for the test

Conclude and estimate the remaining model parameters (depending on results of above tests)